

Smart_Agri_API_Test

POST Register a new user

http://127.0.0.1:5001/api/users/register

Body raw (json)

json

```
{
  "username": "test_farmer_9",
  "password": "password123",
  "email": "test9@farm.com"
}
```

GET Verify Email (NEW REQUEST)

http://127.0.0.1:5001/api/users/verify/

POST Register duplicate username

http://127.0.0.1:5001/api/users/register

Body raw (json)

```
json

{
  "username": "test_farmer_9",
  "password": "password123",
  "email": "test9@farm.com"
}
```

POST Login as farmer (save token!)



http://127.0.0.1:5001/api/login

AUTHORIZATION Basic Auth

Username <username>

Password <password>

POST Login as admin (save token)



http://127.0.0.1:5001/api/login

AUTHORIZATION Basic Auth

Username <username>

Password

<password>

POST Create farm (farmer token)



http://127.0.0.1:5001/api/farms

AUTHORIZATION Bearer Token

Token

Body raw (json)

json

```
{
  "farm_name": "My Test Farm",
  "crop_type": "Wheat",
  "address": { "area_name": "Belfast", "postcode": "BT7 1NN" },
  "location": { "type": "Point", "coordinates": [-5.9301, 54.5973] }
}
```

GET Get single farm



http://127.0.0.1:5001/api/farms/

AUTHORIZATION Bearer Token

Token

PUT Update MY farm (farmer token)



http://127.0.0.1:5001//api/farms/

AUTHORIZATION Bearer Token

Token

Body raw (json)

```
json

{
  "crop_type": "Barley"
}
```

POST Add a sensor



http://127.0.0.1:5001/api/farms//sensors

AUTHORIZATION Bearer Token

Token

Body raw (json)

json

```
{
  "sensor_id": "SEN-TEST-01",
  "type": "Soil Moisture",
  "status": true,
  "readings": [
    {
      "timestamp": "2026-03-12T12:00:00Z",
      "value": 15.5
    }
  ]
}
```

POST Sync weather (owner only)



http://127.0.0.1:5001/api/farms//sync_weather

AUTHORIZATION Bearer Token

Token

Body raw

```
{
  "farm_name": "My Test Farm",
  "crop_type": "Wheat",
  "address": { "area_name": "Belfast", "postcode": "BT7 1NN" },
  "location": { "type": "Point", "coordinates": [-5.9301, 54.5973] }
}
```

GET Farm insights (owner only)



http://127.0.0.1:5001/api/farms//insights

AUTHORIZATION Bearer Token

Token

GET Irrigation check (owner only)



http://127.0.0.1:5001/api/farms//irrigation_check

AUTHORIZATION Bearer Token

Token

PUT Try to update SOMEONE ELSE'S farm (farmer token)



http://127.0.0.1:5001/api/farms/11111111111111111111111111111111

AUTHORIZATION Bearer Token

Token

Body raw (json)

json

```
{ "crop_type": "Barley" }
```

POST Try adding sensor to someone else's farm



http://127.0.0.1:5001/api/farms/11111111111111111111111111111111/sensors

AUTHORIZATION Bearer Token

Token

Body raw (json)

```
json

{
  "sensor_id": "SEN-TEST-01",
  "type": "Soil Moisture",
  "status": true
}
```

POST Sync weather on someone else's farm



http://127.0.0.1:5001/api/farms/11111111111111111111111111111111/sync_weather

AUTHORIZATION Bearer Token

Token

DELETE Delete farm as farmer (should fail)



http://127.0.0.1:5001/api/farms/

AUTHORIZATION Bearer Token

Token

GET Text search

http://127.0.0.1:5001/api/farms/search?q=Belfast

PARAMS

q	Belfast
---	---------

GET Regional insights (public)

http://127.0.0.1:5001/api/farms/region/Belfast/insights

POST Broadcast as farmer



http://127.0.0.1:5001/api/farms/alerts/broadcast

AUTHORIZATION Bearer Token

Token

Body raw (json)

```
json

{
  "alert_type": "Flood",
  "danger_zone": {
    "type": "Polygon",
    "coordinates": [[
      [-7.5, 54.0],
      [-5.5, 54.0],
      [-5.5, 55.0],
      [-7.5, 55.0],
      [-7.5, 54.0]
    ]]
  }
}
```

POST Broadcast alert as admin



<http://127.0.0.1:5001/api/farms/alerts/broadcast>

AUTHORIZATION Bearer Token

Token

Body raw (json)

json

```
{
  "alert_type": "Flood",
  "danger_zone": {
    "type": "Polygon",
    "coordinates": [[
      [-7.5, 54.0],
      [-5.5, 54.0],
      [-5.5, 55.0],
      [-7.5, 55.0],
      [-7.5, 54.0]
    ]]
  }
}
```

DELETE Delete farm as admin



http://127.0.0.1:5001//api/farms/

AUTHORIZATION Bearer Token

Token

GET Logout



http://127.0.0.1:5001/api/logout

AUTHORIZATION Bearer Token

Token

POST Create farm with no token



http://127.0.0.1:5001/api/farms

AUTHORIZATION Bearer Token

Token <token>

Body raw (json)

```
json

{
  "farm_name": "My Test Farm",
  "crop_type": "Wheat",
  "address": { "area_name": "Belfast", "postcode": "BT7 1NN" },
  "location": { "type": "Point", "coordinates": [-5.9301, 54.5973] }
}
```

POST Login with wrong password



http://127.0.0.1:5001/api/login

AUTHORIZATION Basic Auth

Username <username>

Password <password>

GET Get all farms (public)

<http://127.0.0.1:5001/api/farms>

GET Get farm with fake ID

<http://127.0.0.1:5001/api/farms/11111111111111111111111111111111>

GET Text search 1

<http://127.0.0.1:5001/api/farms/search?q=BT7>

PARAMS

q	BT7
---	-----

GET Text search 2

<http://127.0.0.1:5001/api/farms/search>

GET Broadcast alert verify

<http://127.0.0.1:5001/api/farms/69b27c9d974c8a3c29673ea8>